Quiz 9

C6:

(A) $\int (x+2)\ln(x)\,dx$

$= \int x\ln(x)\,dx + \int 2\ln(x)\,dx$

$= \int x\ln(x)\,dx + 2(1/x) + C$

Let $u = x$ | $dv = \ln(x)\,dx$

$du = dx$ | $v = 1/x + C$

$\int u\,dv = uv - \int v\,du$

$= x(1/x) - \int \frac{1}{x}\,dx + \frac{2}{x} + C$

$$\boxed{\int (x+2)\ln(x)\,dx = 1 - \ln|x| + \frac{2}{x} + C}$$

(B) $\int x^2 \sin x\,dx$

u	dv
x^2 +	$\sin x$
$2x$ −	$-\cos x$
2 +	$-\sin x$
0	$\cos x$

$\int x^2 \sin x\,dx$

$= x^2(-\cos x) - 2x(-\sin x) + 2(\cos x) + C$

$= -x^2\cos x + 2x\sin x + 2\cos x + C$

$$\therefore \boxed{\int x^2 \sin x\,dx = -x^2\cos x + 2x\sin x + 2\cos x + C}$$

C7: $\int \tan^3 x \sec^2 x \, dx = \int \tan^2 x \sec^2 x \tan x \, dx$

$\tan x = u$
$du = \sec^2 x$

Let $u = \tan x$, $du = \sec^2 x \, dx$

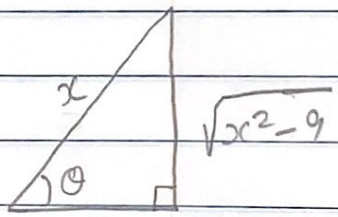
$= \int u^3 \, du = \frac{u^4}{4} + C$

$= \frac{\tan^4 x}{4} + C$

$\therefore \boxed{\int \tan^3 x \sec^2 x \, dx = \frac{\tan^4 x}{4} + C}$

C9: $\int \frac{\sqrt{x^2-9}}{x^3} \, dx$

$x = a \sec\theta$
$x = 3 \sec\theta$
$dx = 3 \sec\theta \tan\theta \, d\theta$

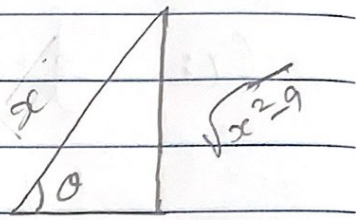


$= \int \frac{\sqrt{9\sec^2\theta - 9}}{27 \sec^3\theta} \, 3 \sec\theta \tan\theta \, d\theta$

$= \int \frac{3 \tan\theta}{9 \sec^3\theta} \sec\theta \tan\theta \, d\theta$

$= \frac{1}{3} \int \frac{\tan^2\theta}{\sec^2\theta} \, d\theta = \frac{1}{3} \int \frac{\sin^2\theta}{\cos^2\theta} \cdot \left(\frac{\cos^2\theta}{1}\right) d\theta$

C8: $= \frac{1}{3}\int \sin^2\theta \, d\theta$



$= \frac{1}{3}\int \left(\frac{1-\cos 2\theta}{2}\right) d\theta$

$= \frac{1}{6}\int 1 \, d\theta - \frac{1}{6}\int \cos 2\theta \, d\theta$

$= \frac{1}{6}\theta - \frac{1}{6}\left(\frac{\sin 2\theta}{2}\right) + C$

$= \frac{1}{6}\theta - \frac{1}{6}\left(\frac{2\sin\theta\cos\theta}{2}\right) + C$

$= \frac{\theta}{6} - \frac{1}{6}\sin\theta\cos\theta + C$

Sub:-

$\theta = \sec^{-1}(x/3)$ and in term of x

$= \frac{\sec^{-1}(x/3)}{6} - \left(\frac{1}{6}\right)\left(\frac{\sqrt{x^2-9}}{x}\left(\frac{3}{x}\right)\right) + C$

$= \frac{1}{6}\left[\sec^{-1}(x/3) - \frac{3\sqrt{x^2-9}}{x^2}\right] + C$

$$\therefore \boxed{\int \frac{\sqrt{x^2-9}}{x^3}\,dx = \frac{1}{6}\left[\sec^{-1}(x/3) - \frac{3\sqrt{x^2-9}}{x^2}\right] + c}$$